



Novel synergistic activities of tetracycline copper oxide nanoparticles integrated into chitosan micro particles for delivery against multiple drug resistant strains: Generation of reactive oxygen species (ROS) and cell death

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ABSTRACT

One of the main strategies on the antibiotic resistance is the synergy between antibiotics and nanoparticles either free or integrated into carriers and slow release. Copper oxide nanoparticles (CuONp) and chitosan microparticles (CsMp) were synthesized by chemical precipitation and ionic gelation method, respectively. CuONp was integrated into the CsMp by sonication method and denoted as CsMp@CuONp. The results revealed the uniform distribution of copper oxide nanoparticles in chitosan matrix with an average size of 8 nm and 1 μm for CuONp and CsMp, respectively. The antibacterial and blood compatibility of CuONp and the synergistic studies of tetracycline/CuONp and tetracycline/CsMp@CuONp were investigated. The average fractional inhibitory concentration (FIC) of tetracycline with CuONp was 0.85 and for tetracycline with CsMp@CuONp reduced to 0.44 which indicated the additive and synergic effect respectively. The generation of reactive oxygen species (ROS) as the primary cause of nanotoxicity and bacterial cell death were achieved by 30 ppm and 60 ppm copper oxide nanoparticles for Gram positive and Gram negative strains respectively. CsMp@CuONp has a greater synergic effect for antibacterial activity of tetracycline compared with CuONp. The CuONp shows interaction with antibiotic and act as tetracycline carrier with enhanced its accumulation interacted with bacteria. The integration of CuONp into the CsMp resulted in blood compatibility at concentrations below 60 ppm.

1. Introduction

Application of antibacterial agents in composition form, blended type with nanoparticles and medicine carriers are new effective pathways in overcoming ever growing resistance of medicines [1,2]. Copper oxide nanoparticles shows antimicrobial characteristics and wound restoration properties with anti-microbial effect through numerous biological paths and low bacterial resistance against [1]. Chitosan is a biologically compatible biopolysaccharide with anti-microbial characteristic and low toxicity have been utilized as medicine carrier [3].

In this research for the first time composites of tetracycline (tet) with copper oxide nanoparticles (CuONp) and copper oxide nanoparticles integrated in chitosan micro-particles (CsMp@CuONp) as antibacterial elements are employed. The synergic effect of copper oxide nanoparticles with tetracycline antibiotics and copper oxide nanoparticles integrated in chitosan micro-particles is reported for the first time.

2. Experimental

2.1. Materials

Copper nitrate trihydrate, $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$, sodium hydroxide and absolute ethanol were purchased from Merck (Germany). Tetracycline hydrochloride (tet) was purchased from Sigma (USA). Antibiotic disks were purchased from Padtan TEB Company (Iran). Chitosan (medium molecular weight, 150–200 kDa, degree of deacetylation 75–90%) and glacial acetic acid and were purchased from Sigma (USA). Deionized water was used for all experiments.

2.2. Bacterial strains

Multidrug-resistant (MDR) coagulase negative *Staphylococcus*, MDR *Pseudomonas* p41, MDR *Pseudomonas* p21, *Staphylococcus aureus* ATCC 6538 and *Pseudomonas aeruginosa* ATCC 9027 were donated by department of microbiology, school of medicine, Tehran University of Medical Sciences.

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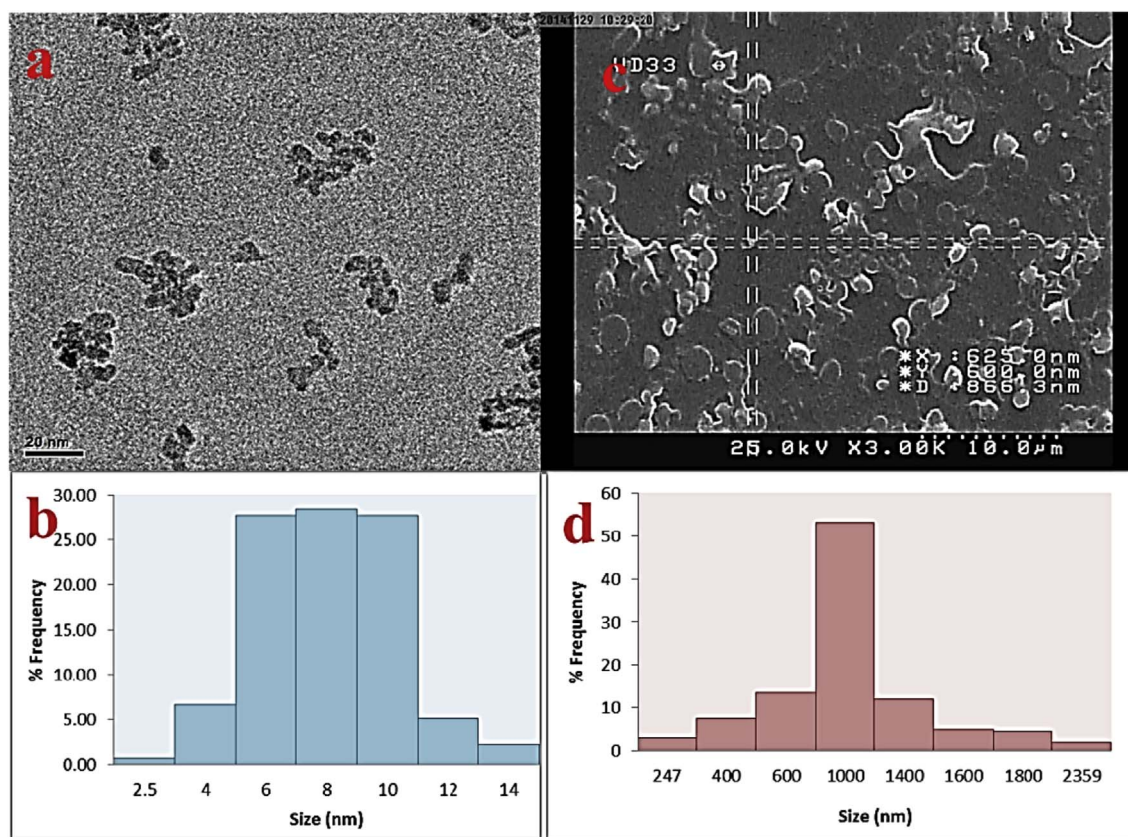


Fig. 1. (a, b) TEM image and size distribution of copper oxide nanoparticles, (c, d) FESEM image and size distribution of chitosan microparticle.

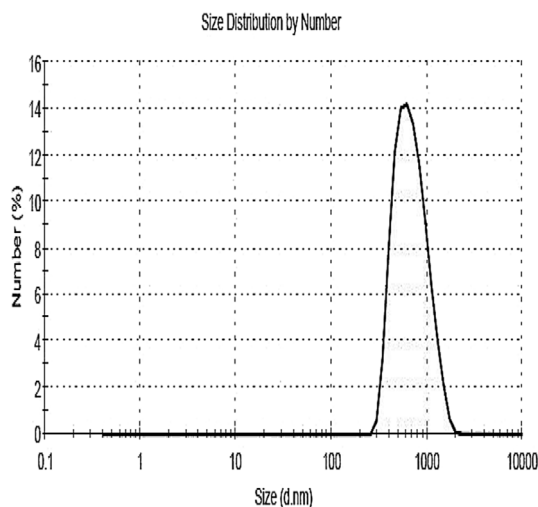


Fig. 2. Particle size distribution of chitosan microparticle decorated by copper oxide nanoparticles by DLS.

2.3. Synthesis of copper oxide nanoparticles

Copper oxide nanoparticles were synthesized by a modified chemical precipitation method. In brief, 90 mg $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ at the concentration of 0.02 M was heated to 80 °C under reflux with addition of 400 μl 1 M NaOH solution. The color of the solution turned to dark blue, 157 mg platelet NaOH was rapidly added to the solution to reach the final pH of 5–6. The black precipitate of CuO nanoparticle were centrifuged, washed and dried in the vacuum oven.

2.4. Preparation of chitosan microparticles

Chitosan powder was dissolved in diluted acetic acid overnight at (0.05%). The solution was filtered by 0.45 μm filter Millipore, the pH of solution was adjusted to 4.7 and microparticles were prepared through ionic gelation method by means of tri-polyphosphate (TPP). 400 μl of TPP 0.3% was added to the mixture as a cross linker by a drop wise method as the solution was mixing. The synthesis of microparticles was immediate due to the strong ionic interactions between phosphate groups in TPP and amine groups in chitosan [4,5].

2.5. Preparation of chitosan microparticles decorated with copper oxide nanoparticles

Copper oxide nanoparticles (0.5 mg) were dispersed in 1 ml water by sonication. Then 500 μl of precipitate of the chitosan microparticle solution was added to the colloidal solution of copper oxide nanoparticles and was sonicated for 20 min. The resulting composites were centrifuged at 10000 rpm for 20 min.

2.6. Characterization of particles

The morphologies and size were characterized by the transmission electron microscope (TEM, Tecnai 20 Model) and field emission scanning electron microscope (FE-SEM, HITACHI S-4160 Model). The average particle size was calculated by dynamic light scattering (DLS, Malvern). The UV–visible spectra were obtained in the region of 200–600 nm (spectrophotometer, Biochrom, England).

2.7. Measurement of CuONp adsorption efficiency

The concentration of copper oxide nanoparticles in supernatant was measured by UV spectroscopy at 280 nm. The absorption efficiency

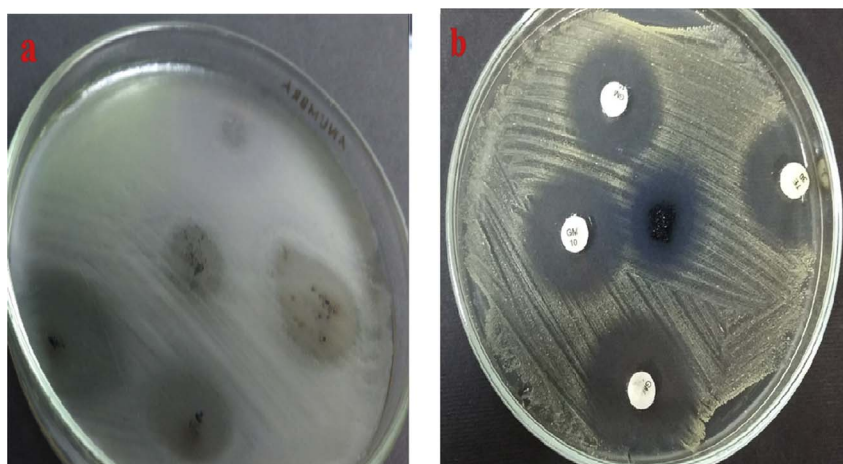


Fig. 3. The inhibition zones of dry CuO nanoparticles against (a) *P.aeruginosa* and (b) *S.aureus*.

Table 1

MICs (μgml^{-1})_a, FIC_b and FIC mean _c and effect of tetracycline in combination with CuONp against resistant isolates and standard bacteria.

Tet/CuONp	<i>S. aureus</i> (ATCC 6538)			MDR coagulase-negative <i>Staphylococcus</i>			<i>P.aeruginosa</i> (ATCC 9027)			MDR <i>Pseudomonas</i> p41			MDR <i>Pseudomonas</i> p21		
	MIC Tet	MIC CuONp	FIC	MIC Tet	MIC CuONp	FIC	MIC Tet	MIC CuONp	FIC	MIC Tet	MIC CuONp	FIC	MIC Tet	MIC CuONp	FIC
Tet alone	0.05	0	–	12	0	–	12	0	–	6.2	0	–	24	0	–
Tet + CuONp	0.025	15	1.03	3	425	0.88	6	1000	1.03	3	500	0.95	12	215	0.64
Tet + CuONp	0.015	20.5	0.98	1.5	534	0.86	3	1250	1.12	1.5	713	0.98	6	350	0.48
CuONp alone	0	30	–	0	700	–	0	1600	–	0	1000	–	0	1500	–
FIC mean			1			0.87			1.07			0.96			0.56
Effect			A			A			I			A			A

_a, _b and _c is average value.

Table 2

MICs (μgml^{-1})_a, FIC_b and FIC mean _c and effect of tetracycline in combination with CsMp@CuONp against resistant isolates and standard bacteria.

Tet C-O	<i>S. aureus</i> (ATCC 6538)			MDR coagulase-negative <i>Staphylococcus</i>			<i>P.aeruginosa</i> (ATCC 9027)			MDR <i>Pseudomonas</i> p41			MDR <i>Pseudomonas</i> p21		
	MIC Tet	MIC C-O	FIC	MIC Tet	MIC C-O	FIC	MIC Tet	MIC C-O	FIC	MIC Tet	MIC C-O	FIC	MIC Tet	MIC C-O	FIC
Tet alone	0.05	0	–	12	0	–	12	0	–	6.2	0	–	24	0	–
Tet + C-O	0.012	7.8	0.45	3	96.2	0.53	6	392	0.74	1.55	196	0.45	3	98	0.19
Tet + C-O	0.006	6.13	0.33	1.5	392	0.68	–	–	–	0.77	392	0.52	0.77	392	0.29
C-O alone	0	30	–	0	700	–	0	1600	–	0	1000	–	0	1500	–
FIC mean			0.39			0.6			0.74			0.48			0.24
Effect			S			A			A			S			S

_a, _b and _c is average value _d: CsMp@CuONp:C-O.

(AE) was calculated using equation (1):

$$\text{Adsorption Efficiency (AE\%)} = \frac{\text{Total CuONp} - \text{Free CuONp}}{\text{Total CuONp}} \times 100 \quad (1)$$

2.8. Antibacterial and synergy assays

2.8.1. Agar diffusion method

The overnight bacterial suspension in the appropriate medium was diluted in PBS to OD₆₀₀ = 0.08 ($\sim 10^7$ CFUml⁻¹) and 100 μ l were inoculated onto 4 mm deep Muller Hinton Agar plate by L929 shape spreader. The dry copper oxide nanoparticles and standard antibiotic disks placed directly onto agar. The inhibition zone of agents was measured separately.

2.8.2. Broth microdilution method

The minimum inhibition concentration (MIC) was carried out on the basis of CLSI (Clinical & Laboratory Standard Institute) in the 96-well polystyrene microtitre plates (4,5). Briefly 100 μ l fresh Mueller Hinton Broth was poured into each well. Then 100 μ l of the 4x concentration of the antibacterial agent were added into the first well, mixed and 100 μ l and transferred to the next well and it repeated. 100 μ l from the final well was discarded and 180 μ l fresh medium was added into each wells. The overnight bacterial suspension in the appropriate medium was diluted in PBS to OD₆₀₀ = 0.08 ($\sim 10^7$ CFUml⁻¹) and 20 μ l inoculated into each well so that each well contained approximately 3×10^4 CFU (5×10^5 CFUml⁻¹). After incubation time (37 °C for 24 h), the concentration of the first well that did not show any visible growth was reported as the MIC value. Checkerboard arrays were used in synergy assays [6]. In standard checkerboard arrays, the MIC of combination of the two antibacterial agents in microtiter wells with several dilutions ($\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$ MIC) of the agents with the same volume was tested.

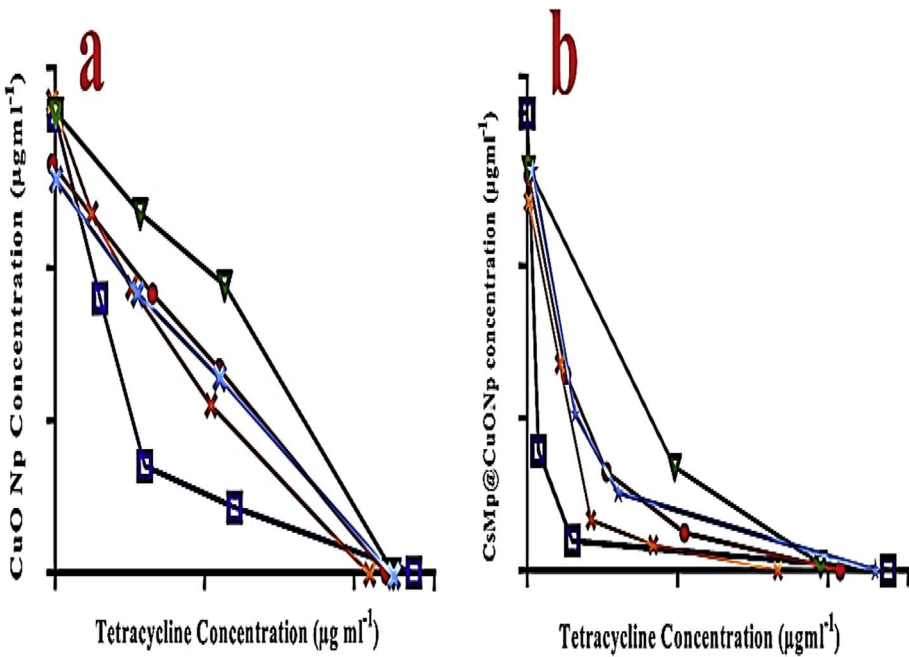


Fig. 4. The isobologram curves of (a) tetracycline with CuONp and (b) tetracycline with CsMp@CuONp against (□) MDR *Pseudomonas* p21, (×) MDR coagulase-negative *Staphylococcus*, (•) *S. aureus* (ATCC 6538), (*) MDR *Pseudomonas* p41 and (▼) *P.aeruginosa* (ATCC 9027).

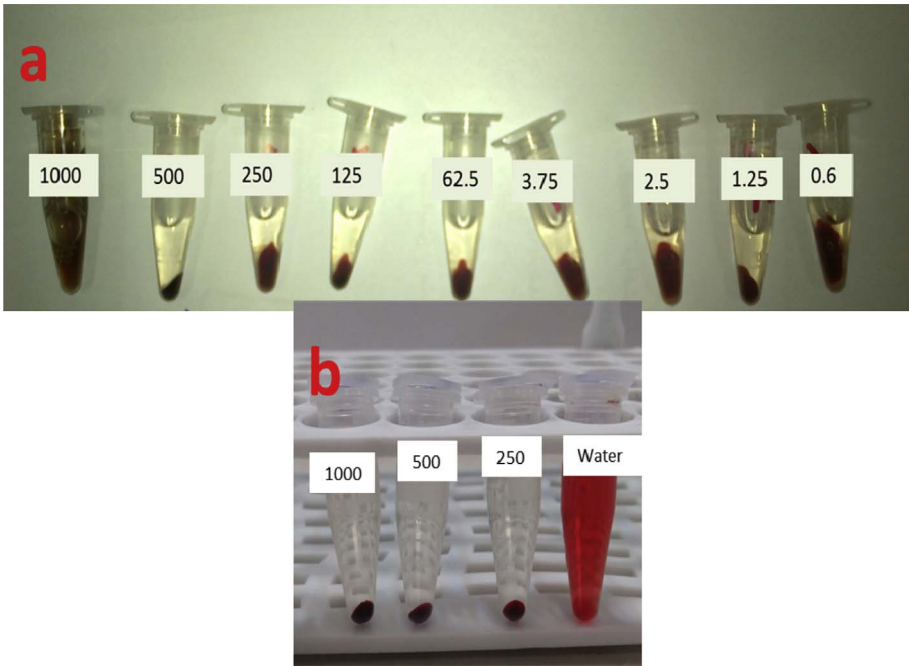


Fig. 5. The hemolysis assay of (a) CuONp and (b) CsMp@CuONp at different concentration and water for positive control.

Table 3
The fluorescence intensity (%) of different concentration of CuONp at 30 min and 60 min exposure to the *St.aureus* compare to positive control (3% H₂O₂).

Concentration (µgml ⁻¹)/time	30 min	60 min
0	10 ± 2	13 ± 1
10	10 ± 3	15.5 ± 2
30	32 ± 0	54 ± 1
90	51 ± 4	70 ± 2
3% H ₂ O ₂	80 ± 2	90 ± 3

Moreover, each agent alone was also tested without the second agent. The Fractional Inhibitory Concentrations (FIC) index and isobologram curves were used for data analyses of checkerboard arrays. FIC indices are quantitative criterion for the prediction of the extent of drug

interaction at a given effective level. The FIC index was calculated based on equation (2):

$$FIC = FICA + FICB = \left(\frac{MICA \text{ in combination}}{MICA} \right) + \left(\frac{MICB \text{ in combination}}{MICB} \right) \quad (2)$$

The synergy was explained as an FIC index of 0.5 or less and additive as an FIC index between of 0.5 and 1.0 and indifferent as an FIC index between 1 and 2.0. Antagonism has been defined as an FIC index More than 2.0 [7]. The isobologram curves fit a graphical presentation of two drugs interaction. In order to draw the isobole curve, the MIC of two agents' combination that inhibited the growth was placed on the x and y-axes. The isobole graph was constructed by connecting the points. Synergy, additive or antagonism was denoted by the degree of

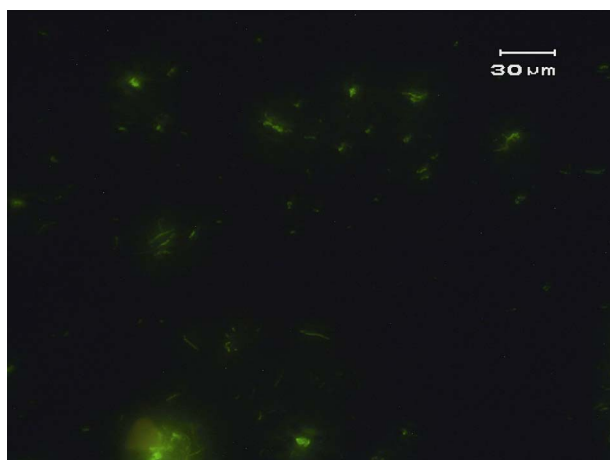


Fig. 6. Destruction of *Staphylococcus* in present of copper oxide nanoparticles detected by florescent microscope.

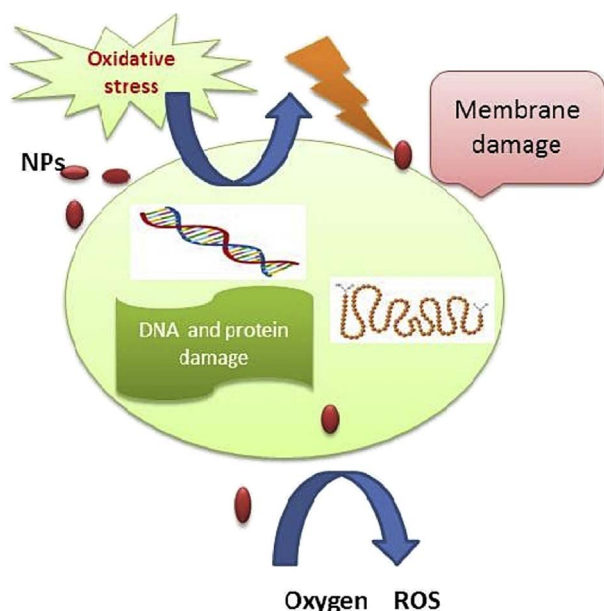


Fig. 7. Overview toxicity of copper oxide nanoparticles and anti-bacterial activities.

curvature of the isobole graph. For a better view of the curvature, the diagonal line of additive can be drawn.

2.9. Hemolysis assay

Citrated blood (3 ml) was washed three times with phosphate buffer, centrifuged, and the supernatant was separated from pellets. The 200 ml of washed blood was completely mixed with 4 ml of phosphate buffer. Washed diluted dilutions (200 μ l) were added to 800 μ l of different dilution of the sample and were vortexed slowly. Pure water and phosphate buffer was used as a positive and negative control, respectively. The tubes kept at room temperature for 2 h and then centrifuged at 1600 for 5 min. The supernatant fluid was read at 541 nm by spectrophotometric to determine the hemolysis. The percentage of hemolysis was calculated from the following equation (3) [8]:

Percent Hemolysis

$$= \frac{\text{Sample absorbance} - \text{Negative control absorbance}}{\text{Positive control absorbance} - \text{Negative control absorbance}} \quad (3)$$

2.10. Oxidative stress assay

Detection of intracellular total ROS were performed using dichlorodihydrofluorescein-diacetate (DCFHDA). Briefly, overnight culture of *S. aureus* was transferred to a 48 well plate at the concentration of 10^7 cells/ml and exposure with the CuONp at the concentration of 0, 10, 30 and 90 ppm. H_2O_2 3% was used as a positive control. The plate was incubated in 37 $^\circ\text{C}$. After 30 and 60 min, the cell suspension centrifuged at 4000 g for 10 min and the pellet was washed with PBS. Then DCFH-DA was added to a final concentration of 60 μM . Water was exposed to dye as negative control. After 30 min incubation at 37 $^\circ\text{C}$ in the dark, 200 μ l of cell suspension was made up to 1 ml deionized water and ROS was assessed by immediately analyzing the cells by F-93 fluoro-spectrophotometer at an excitation wavelength of 488 nm and emission wavelength of 535 nm. The fluorescence intensity (%) was calculated according to the following equation:

$$\text{Fluorescence (\%)} = 100 \times (\text{fluorescence unit sample}) / (\text{fluorescence unit induced by } \text{H}_2\text{O}_2)$$

2.11. The fluorescence microscopy assay

An overnight *S. aureus* was centrifuged and washed with PBS. The pellet was suspended in PBS and exposed with DCFDA at concentration of 100 $\mu\text{g}/\text{ml}$ and CuO nanoparticle for 15 min. Then the cells were centrifuged and washed with PBS again. The resuspended cells were examined by fluorescent microscope.

3. Results and discussion

Electron microscopy images and the distribution histogram of copper oxide nanoparticles and chitosan microparticle are shown in Fig. 1. The copper oxide nanoparticles showed an average size of 8 nm with narrow size distribution mainly spherical. The spherical CsMp were prepared with an average size of 1 μm and normal size distribution (Fig. 1). The average hydrodynamic size and size distribution of chitosan microparticle decorated with copper oxide nanoparticles was determined using DLS measurements (Fig. 2). The absorption efficiency of CuONp onto the CsMp was 78.48 ± 0.24 . Impregnated disk by CuO nanoparticles, direct addition of nanoparticle suspension onto agar and dry nanoparticle implantation onto agar were used for agar diffusion to evaluate the antibacterial activity of CuO nanoparticles against *P. aeruginosa* and *S. aureus*. The dry implantation method showed an inhibition zone among the three methods. The inhibition zone of CuONp against *P. aeruginosa* and *S. aureus* was 18 ± 3 and 17 ± 1 , respectively and with the respect of $p < 0.05$, there is no significant difference (Fig. 3). The MIC of tetracycline in comparison with CuONp (Table 1) and with CsMp@CuONp (Table 2) and the FIC indices against the standard strains as well as the resistant isolates were measured through microdilution assays. In the MIC tests, there were differences in susceptibility and Gram-positive bacteria were more susceptible than Gram-negative ones. Moreover, MDR coagulase-negative *Staphylococcus* was more susceptible to CuONp among the resistant isolates. CsMp at using concentrations did not show any antibacterial effect in agar and broth media. The antibacterial effect of tetracycline in combination of CuONp was especially strong against MDR *Pseudomonas* p21 (Table 1). The antibacterial activity was shown at CuONp concentrations of MICCuO/5 and MICCuO/7, and the MICs of tetracycline being as much as 8-fold lower. The antibacterial effect of tetracycline in combination of CsMp@CuONp, tetracycline and CsMp @ CuONp and CsMp alone in used concentration have not been able to inhibit any of the bacteria and therefore, synergistic computations have only been made using CuONp concentrations. The results show that tetracycline with CsMp@CuONp in most bacterial strains has synergistic effect ($\text{FIC} \leq 0.5$) and has some cumulative effect. Also, curvature of isobulogram 21*Pseudomonas* MDR

curves is more and more curved (Fig. 4a), MDR coagulase negative *Staphylococcus*, *Pseudomonas P41* and *S. aureus* are the same size and inward. The curvature of *P.aeruginosa* (the standard strain) was inwardly but the inhibition was less than the other strains. The greater curvature showed the degree of synergistic effect (Fig. 4b). The results of hemolysis assay indicate that CuONp is synthesized to a concentration of $\mu\text{g/ml}$ -500 and after stabilization in CsMp, up to a concentration of 1000 $\mu\text{g/ml}$ has no hemolytic effect and is completely compatible with blood (Fig. 5a and b). However the concentration of 60 ppm generates the ROS and causes the cells dead (Table 3) in *Staphylococcus*. The destruction of *Staphylococcus* in higher concentration of copper oxide is shown in Fig. 6. The results of this research indicated that CsMp@CuONp with tetracycline, leads to growth control of all bacterial strains, whether standard or resistant. The CsMp alone did not inhibit bacteria, but it stabilized CuONp on its surface and introduces it to the bacteria. It seems that due to the binding effect of amino groups in chitosan with the anionic membrane of bacteria [9], CuONp has more chance to be interacted with bacterial surfaces. In addition, since the noncovalent bound of tetracycline to copper has been reported in environmental studies [10], such a bound may result in increasing density of antibiotics in the membrane of bacteria and easing off connection of antibiotic and the bacteria. Since synergic effect of copper bound to tetracycline with chitosan has been reported in environmental studies [10], such bounds could justify synergic property of our compound. Also, it seems when CuONp is utilized alone, due to aggregation it is not able to implement its anti-microbial effect in low density. On the other hand, CsMp will result in decreasing toxicity of CuONp in higher densities and increasing blood compatibility. In addition, the results indicate that CuONp only in its solid form in agar media will have antimicrobial effect. Santos has used only dry nano-particles in AGAR environment [11]. It seems that dry nanoparticles in agar are more absorbed to the surface of the bacteria. It also illustrated that sensitivity of positive Gram bacteria to CuONp is more than negative Gram. Azem and his associates have also reported the sensitivity of Positive Gram more than all kind of negative Gram [12]. For the first time, in this research stabilization of CuONp in CsMp and its role in increasing better synergic effect with antibiotics has been presented. The results of this research are also notably important from the point of view that resistant isolates through such method will become sensitive bacteria themselves. Also both in vivo and in vitro studies have shown that nanoparticles are closely associated with toxicity by generating reactive oxygen species (ROS) levels or induced the levels of pro-inflammatory mediators. The DNA, protein and cell membrane disrupted upon ROS induction by nanoparticles. The over view of this phenomenon are shown in Fig. 7. The slow antibiotic delivery device has been manufactured by cotton fabric- PLA [13], chitosan-collagen [14], gelatin-alginate [15], electrohydrodynamic-fiber [16] and composite biodegradable scaffold [17]. However in this work the antibiotic and copper oxide were entrapped in to chitosan and cell death analyzed by ROS production by a good compatibility with blood cell.

4. Conclusion

Copper oxide nanoparticles and chitosan microparticles were synthesized by chemical precipitation and ionic gelation method, respectively. Copper oxide nanoparticles was integrated into the chitosan microparticles and the results revealed the uniform distribution of copper oxide nanoparticles in chitosan matrix with an average size of 8 nm and 1 μm for copper oxide nanoparticles and chitosan microparticles were synthesized, respectively. The antibacterial and blood compatibility of copper oxide nanoparticles and the synergistic studies of tetracycline/copper oxide nanoparticles and tetracycline/

CsMp@CuONp were investigated. The average fractional inhibitory concentration (FIC) of tetracycline with CuONp was 0.85 and for tetracycline with CsMp@CuONp reduced to 0.44 which indicated the additive and synergic effect respectively. The generation of reactive oxygen species (ROS) as the primary cause of nanotoxicity and bacterial cell death were achieved by 30–60 ppm copper oxide nanoparticles. CsMp@CuONp has a greater synergic effect for antibacterial activity of tetracycline compared with CuONp. The CuONp shows interaction with antibiotic and act as tetracycline carrier with enhanced its accumulation interacted with bacteria. The integration of CuONp into the CsMp resulted in blood compatibility at concentrations below 60 ppm.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jddst.2017.11.017>.

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